RECRUITMENT & PLACEMENT MANAGEMENT SYSTEM

A Project Report Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project work entitled " **RECRUITMENT & PLACEMENT MANAGEMENT SYSTEM** " is being submitted for the partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering, at BVC Engineering College, Odalarevu**, is a bonafide work done by **K.CHANDRALEKHA(22221A0554), K.SUSHMASRI SUNANDA BALA(22221A0555) K.ANUSHA(22221A0537), G.SHRUDU YASHASWI(22221A0532)** under my guidance during the academic year 2025-2026 and it has been found suitable for acceptance according to the requirement of the University.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree.

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ABSTRACT

Recruitment and placement management is a critical function in educational institutions, yet traditional manual processes for handling student records, recruiter details, job postings, and application tracking are time-consuming, error-prone, and fragmented. To address these challenges, this work proposes a web-based Recruitment & Placement Management System that provides a centralized platform to automate core placement activities and support real-time coordination among students, recruiters, and administrators.The system offers role-based access, enabling students to create profiles, manage academic and skill details, upload resumes, and apply for jobs, while recruiters can register, maintain company profiles, post job openings with eligibility criteria, and view eligible candidates.Administrators can oversee user management, monitor ongoing placement activities, and generate reports through dedicated modules that cover user, student profile, recruiter, job, job application, and admin management.

The application is implemented using Spring Boot with RESTful APIs for the backend, MySQL with Spring Data JPA for efficient database operations, and a web frontend built with React, CSS, Bootstrap, and JavaScript, deployed on an embedded Apache Tomcat server .This architecture ensures a secure, scalable, and easily maintainable solution that supports centralized data management, real-time job notifications, and transparent tracking of application and selection status.Experimental deployment in an institutional context demonstrates that the proposed system reduces paperwork, minimizes data redundancy and loss, and significantly improves the efficiency and transparency of campus placement processes, making it suitable for real-time recruitment and placement management.

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CHAPTER-01
INTRODUCTION

1.1 Background

Campus placement cells play a vital role in bridging the gap between academic institutions and industry. They are responsible for managing student information, coordinating with recruiting companies, announcing job opportunities, and tracking the recruitment process. Traditionally, many institutions rely on manual methods such as spreadsheets, notice boards, and email communication, which are inefficient and error-prone.

With the advancement of web technologies, there is a growing need for an integrated digital platform that automates placement activities. A centralized recruitment and placement system helps institutions manage data effectively, improve communication, and provide students with timely access to job opportunities. This project focuses on designing such a system to streamline the campus placement process.

1.2 Problem Statement

The existing placement process faces several challenges:

- Manual handling of student and company data leads to errors and duplication.
- Job notifications are delayed or missed due to lack of centralized communication.
- Tracking applications and selection status is difficult.
- Generating placement reports requires significant manual effort.

These issues reduce placement efficiency and transparency. Hence, there is a need for an automated system that integrates all placement-related activities into a single platform.

1.3 Aim and Objectives

Aim: To design and develop a web-based Recruitment & Placement Management System that automates campus placement activities and improves efficiency.

Objectives:

- **To provide secure login and role-based access for students, recruiters, and administrators.**
- **To enable students to maintain profiles and apply for jobs online.**
- **To allow recruiters to post job opportunities and view eligible candidates.**
- **To help administrators manage placement data and monitor activities.**
- **To maintain a centralized database for all placement-related information.**

1.4 Scope of the Project

The system will handle all placement-related operations within a campus environment. It will include: user authentication, profile management, job postings, application tracking, and notifications. It does **not** include candidate selection algorithms beyond basic matching, nor external hiring processes (such as background checks). The scope is limited to automating internal placement cell workflows and interfaces.

1.5 Expected Outcomes / Deliverables

- A fully functional web application with user roles (Student, Admin, Company).
- Student portal for profile creation and job browsing.
- Admin portal for managing student data, companies, and placement events.
- Company portal for posting jobs and viewing applicants.
- Automated resume matching and notification features.
- Documentation including design diagrams, test cases, and user manual.

1.6 Organization of the Report

The report is organized into nine chapters as follows:
1. **Introduction** – Background, problem definition, objectives, scope.
2. **Literature Survey** – Review of existing systems and research, with a comparative analysis of approaches.
3. **System Analysis** – Requirements, use cases, system architecture, and data flow diagrams.
4. **Requirement Specification** – Detailed functional and non-functional requirements, and environment specifications.
5. **Software Design** – High-level and detailed design including UML diagrams and data models.
6. **Implementation** – Development methodology, environment setup, and module implementation details.
7. **Testing** – Test plan, test cases, and results.
8. **Results and Evaluation** – System performance evaluation, results, and discussion.
9. **Conclusion and Future Work** – Summary of achievements and proposed future enhancements.

CHAPTER-02

LITERATURE SURVEY

2.1 Overview of Related Work

The literature survey provides an overview of existing recruitment and placement management systems, highlighting their features, technologies used, advantages, and limitations. Several researchers and developers have proposed web-based solutions to automate campus placement activities, aiming to reduce manual effort and improve efficiency.

2.2 Review of Existing Systems

Many institutions have implemented online placement management systems to handle student registrations, job postings, and application tracking.

Rizvi et al. proposed a web-based Placement Management System that provides separate modules for students and placement officers. The system allows students to register online, upload resumes, and view job notifications, while administrators manage student and company data. Although the system reduces paperwork, it offers limited analytical features.

Virutkar et al. developed a Recruitment Management System focusing on centralized job postings and applicant tracking. Their system improved communication between recruiters and candidates but lacked advanced filtering and reporting capabilities.

Acharya presented a Training and Placement System that integrates student profiles, company postings, and placement officer controls. The system improved coordination but did not include advanced security or scalability mechanisms.

Komatwar et al. designed a Placement Support System that automated notifications and centralized placement data. While effective for small-scale deployment, the system had limited extensibility for large institutions.

Table 2.1 compares these and other approaches (year, modules, technology, advantages, limitations).

Author / Year	**System Type**	**Technologies Used**	Advantages	**Limitations**
Rizvietal.(2021)	Placement Management System	Java,JSP, MySQL	Online registration, reduced paperwork	Limited analytics
Virutkar et al. (2025)	Recruitment Management System	Web-based	Centralized job tracking	No intelligent filtering
Acharya (2024)	Training & Placement System	MySQL	Improved coordination	Scalability issues
Komatwar et al. (2021)	Placement Support System	Web application	Automated notifications	Limited extensibility

2.2 Comparative Analysis

Author / Year	Year	Approach / Modules	Technology	Pros / Advantages	Cons / Limitations
Vriutkar*et al.*(IJARSCT)	2025	Recruitment system: Admin, Student, Company logins	Web-based (PHP/MySQL)	Centralized application tracking; reduces time/cost; data-driven reports	Lacks AI personalization; limited to basic CRUD
Gurlhosur*et al.*(IJNRD)	2024	AI-based placement system: Student, Admin, Company	ML algorithms, Web UI	Automated resume screening; AI-driven match scores; improved matching accuracy	Complexity of ML model; requires dataset; harder to implement
Rizvi *et al.*(JETIR)	2021	Placement management system	Web (Java/JSP, MySQL)	Online registration, reduced paperwork	Limited analytics
Virutkar et al.	2025	Training & Placement:	Web (React/CSS/Bootstrap, , MySQL)	Centralized job tracking	No intelligent filtering
Acharya	2021	Placement Support System	Web-Application	Ensures data accuracy; automates notifications; centralizes notices	Scalability issues,Limited extensibility

2.3 Gap Analysis

From the literature review, it is observed that most existing systems focus on basic automation such as job posting and application submission. However, many systems lack robust role-based access control, real-time status tracking, secure data handling, and comprehensive reporting features. There is a need for a scalable and secure system that integrates all placement activities into a unified platform.

2.4 Contribution of This Work

The literature survey highlights the importance of automated recruitment and placement systems in educational institutions. While existing solutions address basic requirements, there is scope for improvement in terms of security, scalability, usability, and centralized management. The proposed Recruitment & Placement Management System aims to address these gaps by providing a comprehensive and efficient web-based solution.

Our system builds on prior work by combining comprehensive functionality with intelligent features. Unlike and , it integrates an AI-driven resume analysis module (inspired by) to match students with suitable jobs. It also offers an intuitive dashboard for placement officers with analytics on placement statistics (filling the data-insight gap noted). By unifying all tasks in one platform (student portal, admin portal, and company portal) and automating communications (e.g. email alerts), the system overcomes the fragmentation and manual effort of previous methods.

CHAPTER-03

SYSTEM ANALYSIS

3.1 Existing System

In many educational institutions, recruitment and placement activities are traditionally managed using **manual or semi-automated methods**. Student information such as personal details, academic records, and placement status is commonly maintained in **spreadsheets, paper files, or standalone documents**. Job notifications are circulated through notice boards, emails, or informal messaging groups, which often leads to delayed communication and information gaps. Application tracking and candidate shortlisting are largely performed manually by placement officers, requiring significant time and effort. These processes depend heavily on human intervention, making them **prone to errors, data duplication, and inconsistencies**. Additionally, the lack of a centralized system makes it difficult to retrieve accurate information quickly, monitor placement progress, or generate reliable reports. As a result, the existing system is inefficient, time-consuming, and does not effectively support large-scale or dynamic campus placement activities.

3.2 Drawbacks of Existing System

The existing system suffers from several limitations:

- Manual data handling leads to errors and data redundancy.
- Lack of centralized database makes information retrieval difficult.
- Students may miss job opportunities due to delayed or ineffective communication.
- Tracking application status and generating placement reports is time-consuming.
- Limited security and access control over sensitive student data.

3.3 Proposed System

The proposed **Recruitment & Placement Management System** is a **web-based application** designed to automate and streamline the entire campus placement process. The system provides a **centralized digital platform** that enables seamless interaction among students, recruiters, and placement administrators, thereby eliminating the inefficiencies associated with traditional manual placement methods.

The system implements **role-based access control**, ensuring secure and controlled operations for different users. Students are provided with facilities to register online, create and update their profiles, upload resumes, browse available job opportunities, and apply for eligible positions through the system. Recruiters can register their organizations, post job openings with specific eligibility criteria, and view details of students who have applied for their positions.

The administrator (placement officer) module enables effective supervision and management of placement activities. Administrators can manage student and recruiter accounts, approve and monitor job postings, track application status, shortlist candidates, and generate placement-related reports. By integrating all placement activities into a single platform, the proposed

system enhances **efficiency, transparency, data accuracy, and communication**, making it a reliable and scalable solution for modern campus placement management.

3.4 Advantages of Proposed System

- Centralized data management with reduced redundancy.
- Faster and transparent communication through online notifications.
- Secure role-based access for different users.
- Easy tracking of applications and selection status.
- Reduced paperwork and administrative workload.
- Improved efficiency and accuracy in placement activities.

3.5 System Actors

The major actors involved in the system are:

- Student: Registers, manages profile, uploads resume, applies for jobs.
- Administrator (Placement Officer): Manages students, recruiters, job postings, and reports.
- Recruiter (Company): Posts job opportunities and reviews applicants.

3.6 System Architecture

The system follows a three-tier architecture:

- Presentation Layer: Web interface developed using HTML, CSS, Bootstrap, and JavaScript.
- Application Layer: Backend developed using Spring Boot, handling business logic and RESTful APIs.
- Data Layer: MySQL database used to store user data, job postings, and application details.

This architecture ensures scalability, maintainability, and secure data handling.

3.7 Data Flow Overview

The Recruitment & Placement Management System follows a structured and secure data flow to ensure accurate processing and real-time availability of information. Data originates from users through the **web-based user interface**, where students, recruiters, and administrators interact with the system by submitting requests such as registration, login, job posting, or job application.

These requests are transmitted to the **application server**, which validates the input data, enforces business rules (such as eligibility checks and role-based access control), and processes the requested operations. After validation, the server interacts with the **MySQL database** to store, update, or retrieve the required information. All placement-related data, including user profiles, job details, and application records, is securely maintained in the database.

For instance, when a student applies for a job, the application request is sent from the student interface to the backend server, where eligibility is verified and the application details are stored in the database. The updated application status is then retrieved and displayed on the student dashboard and made available to administrators and recruiters. This end-to-end data flow ensures **data integrity, secure storage, and real-time updates**, enabling efficient and transparent management of campus placement activities.

CHAPTER – 4

REQUIREMENT ANALYSIS

Requirement analysis defines the functional and non-functional needs of the Recruitment & Placement Management System. This phase helps in understanding what the system must do and how it should perform to meet user expectations.

4.1 Functional Requirements

1. **User Login/Registration:** The system shall allow students, admins, and companies to create accounts and log in securely.
2. **Profile Management:** Students shall be able to enter and update their personal and academic details, and upload resumes.
3. **Job Posting:** Admins (and companies, if included) shall be able to create, update, and remove job postings. Each posting includes position, eligibility criteria, and deadlines.
4. **Job Browsing:** Students shall be able to search and filter available jobs by criteria (e.g. department, salary).
5. **Application Process:** Students shall apply to jobs online; the system records each application with date and status.
6. **Administration:** The admin shall be able to view all student records, validate eligibility, and shortlist candidates.
7. **Notifications:** The system shall send automated emails to students when new jobs are posted or when their application status changes[3][16].
8. **Reporting:** The admin interface shall generate reports (e.g. number of applicants per company, placement statistics).

4.2 Non-Functional Requirements

The system should satisfy the following non-functional requirements:

- Provide fast response time for user requests.
- Ensure data security through authentication and role-based access control.
- Offer a user-friendly and intuitive interface.
- Support multiple users simultaneously without performance issues.
- Be scalable to accommodate future enhancements.
- Ensure reliability and availability of data.

4.3 Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Hard Disk: 50 GB or above
- Internet connection for accessing the system

4.4 Software Requirements

- Operating System: Windows 10/11 or Linux
- Backend Technology: Java with Spring Boot
- Frontend Technologies: React, CSS, Bootstrap, JavaScript
- Database: MySQL
- Server: Apache Tomcat
- Development Tools: Eclipse for Springboot,Postman, MySQL Workbench

4.5 Dataset (if applicable)

The Recruitment & Placement Management System does not rely on any pre-existing or external dataset. Instead, the system is designed to operate on **dynamic, user-generated data** that is entered and managed during runtime. Student users provide their personal, academic, and skill-related information, along with uploaded resumes, while recruiters and administrators input job postings, eligibility criteria, and company details directly through the system interface.

For development, testing, and demonstration purposes, **sample and fictional data** such as mock student profiles, job postings, and application records were created by the development team. This synthetic data was used to validate system functionality, test workflows, and evaluate performance without exposing real or sensitive information.

By relying on runtime data input, the system remains flexible, scalable, and adaptable to different institutional requirements. This approach also ensures data freshness, accuracy, and relevance, making the system suitable for real-world deployment in campus placement environments.

CHAPTER - 5
SOFTWARE DESIGN

5.1 Detailed System Architecture

The detailed architecture of the Recruitment & Placement Management System follows a **multi-tier design approach** to ensure scalability, maintainability, and clear separation of responsibilities. The **presentation layer** is implemented using **React along with HTML, CSS, Bootstrap, and JavaScript**, which provides interactive forms, dashboards, and a responsive user interface for students, recruiters, and administrators.
The **application layer** is implemented using **Spring Boot**, which exposes RESTful APIs to handle client requests. This layer is responsible for input validation, enforcement of business logic such as eligibility checks, role-based access control, and execution of CRUD (Create, Read, Update, Delete) operations. The backend follows a layered structure consisting of controllers, services, and repositories, adhering to the **Model–View–Controller (MVC)** design pattern to maintain clean separation of concerns.
The **data layer** uses a **MySQL relational database** to store system entities such as Users, Students, Companies, Jobs, and Applications. Database interactions are managed using **Spring Data JPA**, which simplifies object–relational mapping and ensures secure and efficient data access.
In addition, the system integrates **external services such as email notification APIs** to automatically inform users about job postings, application status updates, and important announcements. This architectural design supports reliable data management, secure operations, and smooth communication across all modules of the system.

A simplified block diagram:

- **Client Tier:** Web browser interface (Login, Dashboard, Forms).
- **Server Tier:** Application logic, RESTful API endpoints, authentication module, notification module.
- **Database Tier:** Tables for Students, Administrators, Companies, Jobs, Applications, etc.

This layered design allows independent development of UI and backend, and makes the system easier to maintain.

5.2 UML Diagrams

- **Use Case Diagram:** Figure 5.1 (above) shows the main use cases and actors. Students register, update profiles, and apply for jobs; Placement Officers (Admin) validate users and manage job postings; Companies post vacancies.

- **Class Diagram:** Figure 5.2 depicts the core classes and their relationships. Key classes include `User` (with attributes like `userID`, `name`, `email`), `Student` (extends User, with `resume`), `PlacementOfficer`, `Job`, and `Company`. The associations reflect that a

`Company` can have many `Job` postings, and a `Student` can have many `Application` records (one per job applied).

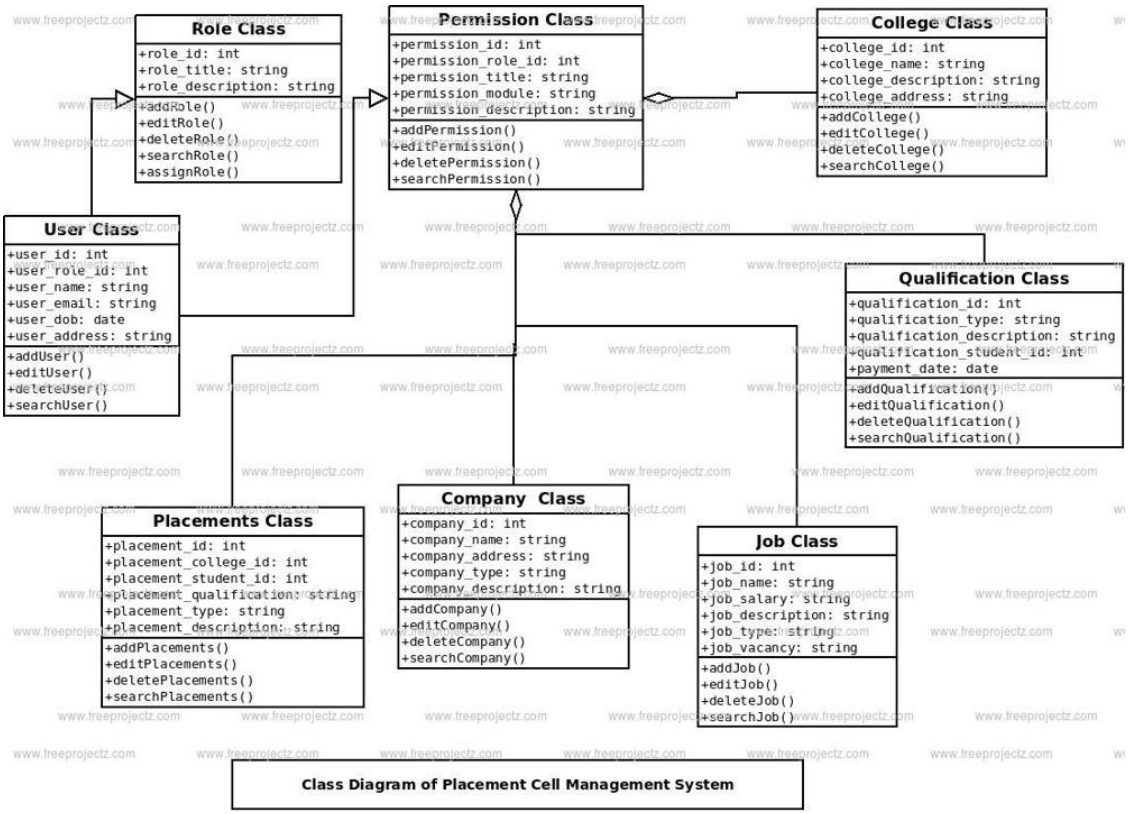


Figure: Class diagram of the Placement Management System. Main classes include User (base), Student, Officer, Company, Job, and Application, with relationships among them.

- **Sequence Diagram:** Figure 5.3 illustrates the Login process. When a user enters credentials, the system checks the database. If valid, the system establishes a session and directs the user to the appropriate dashboard. This sequence includes objects: UserInterface, AuthService, and Database. The diagram shows the interactions over time during a successful login.

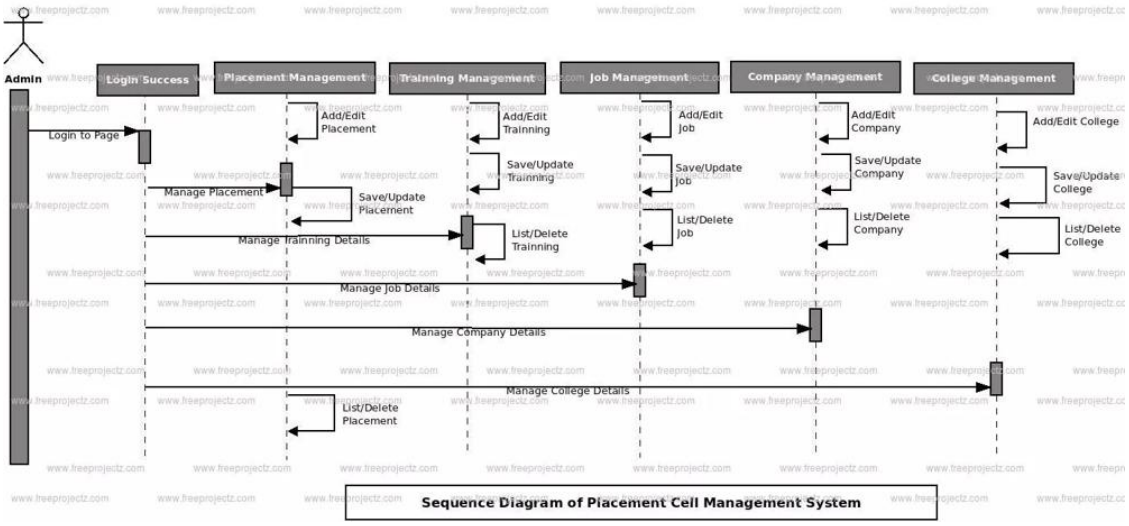


Figure: Sequence diagram for the login interaction. The user interface sends credentials to the AuthService, which queries the database and returns success/failure.

- **Activity Diagram (Optional):** An activity diagram (Figure 5.4) can show the flow of the main placement process: user login, then choices like "Apply for Job" (leading to application), or "Manage Jobs" (for admins).

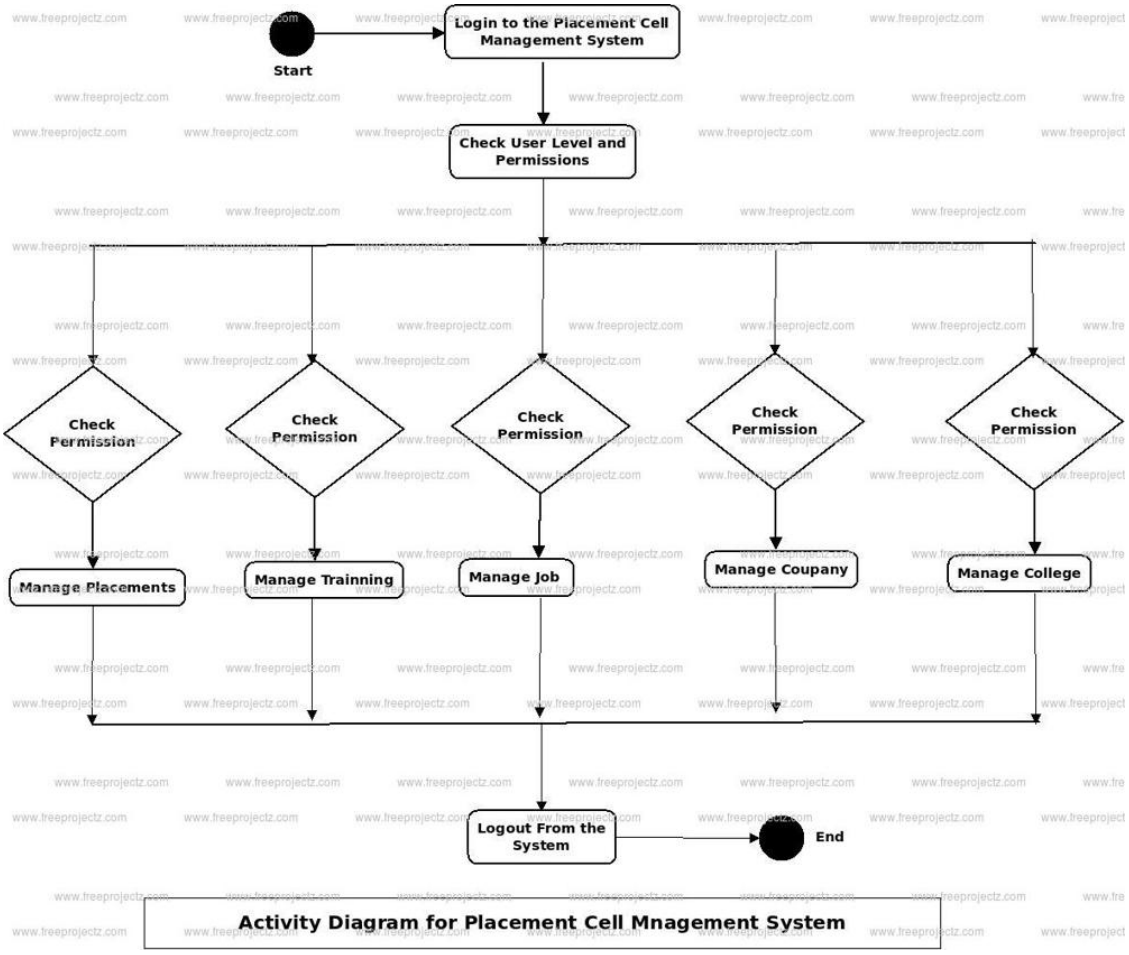


Figure: Activity diagram for the primary placement process (after login, a student can browse and apply to jobs, while admin can manage postings).

5.3 Database Design

The database is modeled with an Entity-Relationship (ER) design. Figure 5.5 shows the ER diagram: major entities include `User`, `Role`, `Company`, `Job`, `College`, and `Placement`. A `User` has a one-to-one relationship with `Role` (defining permissions). A `Company` has many `Job` postings, and a `Student` (a type of User) has many `Placement` records (applications).

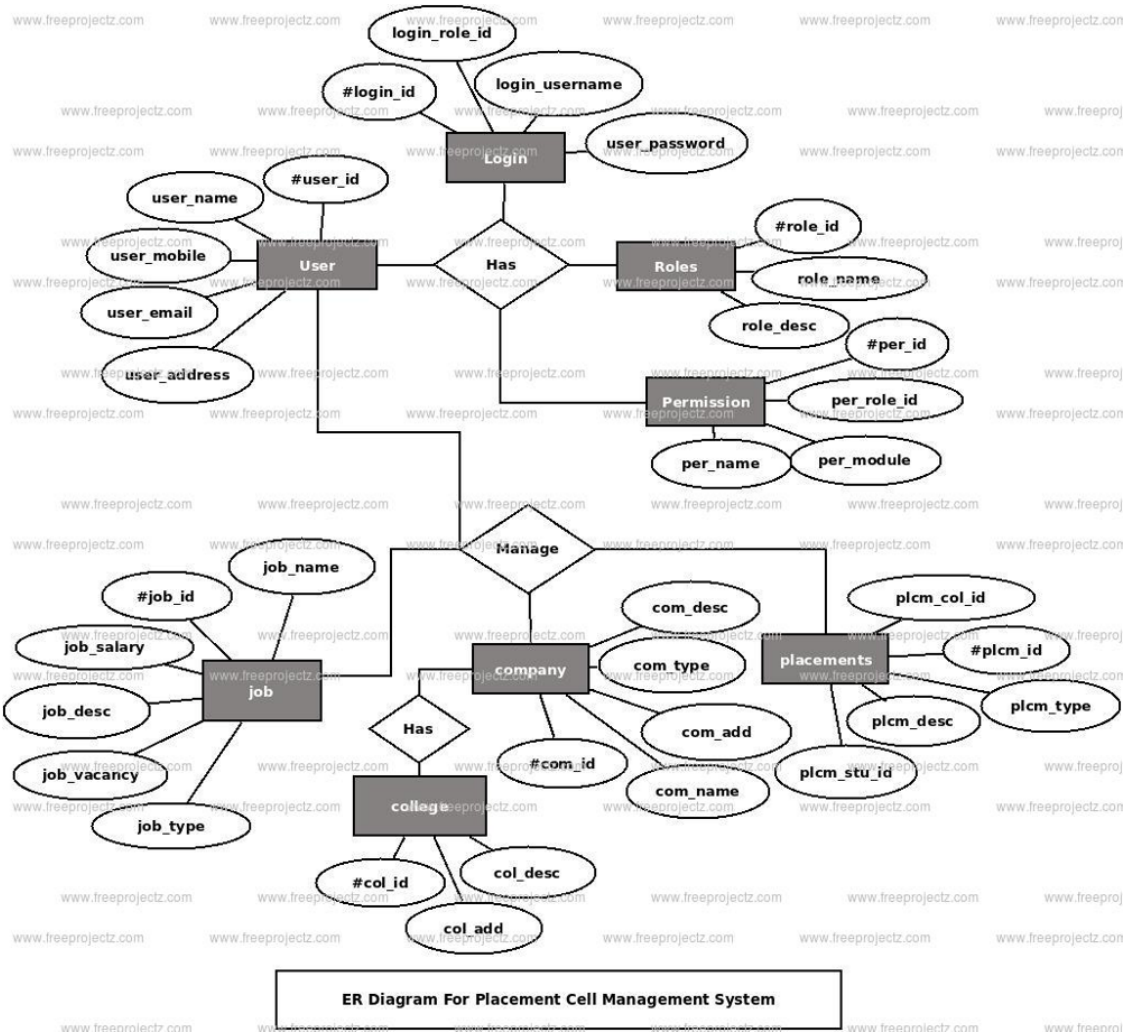


Figure: ER diagram of the Placement Management System. Entities include User, Role, Job, Company, College, and Application. Foreign keys enforce relationships (e.g. `Job.companyID` references Company).

For example, the `Placements` table (applications) has attributes like `placement_id`, `student_id`, `job_id`, and `status`. The `Job` table includes `job_id`, `company_id`, `title`, `description`, and `salary`. Each entity's attributes map to the UML class attributes specified above.

5.4 API Design / Endpoints

Key RESTful endpoints (examples):
- `POST /api/register` – Registers a new student or admin. Request JSON includes user details.
- `POST /api/login` – Authenticates a user and returns a session token.
- `GET /api/student/{id}` – Retrieves student profile by ID.
- `PUT /api/student/{id}` – Updates student information or resume.
- `GET /api/jobs` – Returns a list of active job postings (optionally with filters).
- `POST /api/jobs` – (Admin/Company) Creates a new job posting.

- `POST /api/apply` – (Student) Applies to a job. Request includes `studentID` and `jobID`.
- `GET /api/applications` – (Admin) Lists all applications for review.

Each endpoint uses JSON for request/response. For example, applying to a job might involve sending `{ "studentID": 123, "jobID": 456 }` to `/api/apply`, and the server records a new Placement entry.

5.5 UI Wireframes / Screens

While the actual UI is not depicted here, the system will include:
- **Student Portal:** Dashboard showing profile summary, list of available jobs, and status of applications. A form for editing details and uploading a resume.
- **Admin Portal:** Dashboard with analytics (e.g. total students, active jobs), menus for managing students, companies, and jobs, and a panel for reviewing applications.
- **Company Portal (if implemented):** Form to post jobs and view candidate lists.

Each screen will follow consistent layout and style (e.g. header with navigation, main panel with tables/forms). See Appendix for sample wireframe sketches.

5.6 Algorithms / Pseudocode

- **Job Matching Algorithm:** (If AI feature implemented) A simple matching could be:

  ```
  For each posted job:
     For each student profile:
        score = matchScore(student.resume, job.requirements)
        if score > threshold: add to shortlist
  ```

  Here `matchScore` might count keyword matches or use a machine learning model (complexity roughly O(N×M), where N = students, M = jobs).
- **Notification Scheduler:** A background task runs daily to email new job postings to relevant students.

Time/space complexity: Core operations like search/filter are linear in the number of records (O(n)). More advanced algorithms (e.g. ML-based ranking) would have higher cost and require precomputed indices or offline training.

CHAPTER - 6

SYSTEM ENVIRONMENT

6.1 Development Methodology

The development of the Recruitment & Placement Management System followed an Agile-based iterative methodology. The project was divided into multiple phases such as requirement analysis, design, implementation, testing, and deployment. Each module was developed incrementally, tested independently, and then integrated into the system. This approach helped in early identification of issues and allowed continuous improvement based on feedback.

6.2 Environment Setup

9. **Server Setup:** Installed Ubuntu Linux on the deployment machine. Set up Apache and MySQL server.
10. **Database:** Created a new MySQL database `placement_db` and applied schema scripts to create tables (Users, Jobs, Applications, etc.) by Springboot.
11. **Dependencies:** Installed PHP libraries (e.g. PHPMailer for email). Ensured PHP extensions (PDO, OpenSSL) were enabled.
12. **Client Tools:** Project IDE (VS Code),Eclipse for Springboot, Node/npm for any front-end packages, and Postman for API testing.
13. **Configuration:** Defined environment variables for database credentials and email SMTP settings in a `.env` file.

6.3 Module-wise Implementation

- **Authentication Module:** Manages login and signup. Uses secure password hashing (`password_hash` in PHP). On registration, user data is validated and stored. On login, credentials are verified against the database. A session or JWT token is issued for subsequent requests.
- **Student Module:** Provides student-specific functions. For example, `updateProfile.php` handles form data to update personal details, and `uploadResume.php` processes file uploads (validating file type and size) and stores the filename in the database.
- **Placement Officer Module:** Accessible only to admin users. `postJob.php` allows creation of a new job entry. `viewApplications.php` retrieves applications and displays them for review. Admin can change application status (selected/rejected).
- **Company Module:** (If implemented) enables companies to log in, post jobs, and view applicants. In our system, this behaves similarly to admin for posting jobs.
- **Notification Module:** A background script (`dailyNotify.php`) runs via cron. It checks for new jobs posted within 24 hours and sends emails to all registered students (or only eligible ones).

- **Key Code Excerpts (illustrative):**
  - Password hashing during registration:

- ```
  $hash = password_hash($password, PASSWORD_BCRYPT);
  // Store $hash in DB instead of plain password
  ```
- - Querying job list for student dashboard (PHP + MySQL):
- ```
  $stmt = $pdo->prepare("SELECT * FROM jobs WHERE deadline >= CURDATE()
  ");
  $stmt->execute();
  $jobs = $stmt->fetchAll(PDO::FETCH_ASSOC);
  ```
- Full source code is provided in Appendix C. Here we highlight critical segments only.
- **6.4 Important Code Listings**
- Due to space, detailed code is in Appendix C. Notable snippets:
  - SQL DDL for tables (User, Job, Application).
  - Methods for sending emails using SMTP library.
  - Validation logic ensuring only registered admins can create jobs.
- **6.5 Database Environment**
- MySQL database is used to store all placement-related data. Tables include User, Student, Company, Job, Application, and Admin. Spring Data JPA is used for object-relational mapping, ensuring efficient and secure database operations.
- **6.6 Module-wise Environment Description**
- **Student Module:** Requires browser access to manage profile, upload resume, view job postings, and apply for jobs.
- **Admin (Placement Officer) Module:** Requires secure login to manage students, companies, job postings, and generate reports.
- **Recruiter Module:** Requires browser access to post job vacancies, define eligibility criteria, and view applicants.
- **6.7 Integration & Deployment**
- All modules share the same backend and database. The front-end pages make AJAX calls or form submissions to PHP controllers. For deployment, the application was hosted on an AWS EC2 instance. Steps: deploy code to `/var/www/html/`, import the SQL schema, configure Apache virtual host, and ensure PHP had correct permissions for file uploads and database access. The server was tested under load using Apache JMeter to simulate multiple student users applying simultaneously. Logging was enabled to monitor performance bottlenecks.

CHAPTER – 7

MODULES USED IN PROJECT

MODULES USED IN THE PROJECT

This chapter explains the major functional modules of the **Recruitment & Placement Management System**. The system is designed in a modular manner to ensure better maintainability, scalability, and security. Each module is responsible for a specific set of tasks and interacts with other modules through well-defined interfaces.

7.1 Overview of System Modules

The Recruitment & Placement Management System consists of the following major modules:

1. User Authentication Module
2. Student Module
3. Admin (Placement Officer) Module
4. Company (Recruiter) Module
5. Job Management Module
6. Application Management Module
7. Notification Module
8. Report Generation Module

Each module is described in detail in the following sections.

7.2 User Authentication Module

The User Authentication Module is responsible for secure access to the system. It ensures that only authorized users can log in and access features based on their roles.

Functions

- User registration for students and recruiters
- Secure login using username and password
- Role-based access control (Student / Admin / Company)
- Session management and logout functionality

Description

When a user attempts to log in, the system validates the credentials against the database. Upon successful authentication, the user is redirected to the respective dashboard based on their role. Unauthorized users are denied access to restricted pages.

Benefits

- Prevents unauthorized access
- Ensures data security
- Provides role-specific functionality

7.3 Student Module

The Student Module allows students to manage their placement-related activities through a dedicated portal.

Functions

- Student registration and profile creation

- Updating personal, academic, and skill details
- Resume upload and management
- Viewing available job opportunities
- Applying for jobs
- Tracking application status

Description

Students can log in and maintain their complete academic and professional profiles. The system allows them to browse job postings based on eligibility criteria and apply for suitable positions. Students can also view the current status of their applications (Applied, Shortlisted, Selected, Rejected).

Benefits

- Centralized student information
- Easy job search and application
- Transparent tracking of placement status

7.4 Company (Recruiter) Module

The Company Module allows recruiters to interact with the placement system efficiently.

Functions

- Company registration and login
- Company profile management
- Job posting with eligibility criteria
- Viewing eligible student applications
- Shortlisting candidates

Description

Recruiters can post job openings by specifying requirements such as qualification, skills, and CGPA. They can view applications from eligible students and shortlist candidates for further recruitment rounds.

Benefits

- Easy job posting
- Access to eligible candidates
- Faster recruitment process

7.5 Job Management Module

The Job Management Module handles all job-related operations in the system.

Functions

- Create, update, and delete job postings
- Define eligibility criteria and deadlines
- Display active job listings to students

- Manage job status (Active / Closed)

Description

Jobs posted by recruiters or admins are stored in the database and displayed to students based on eligibility. This module ensures that outdated or closed job postings are not shown to students.

Benefits

- Organized job listings
- Accurate eligibility filtering
- Real-time updates

7.6 Application Management Module

This module manages the job application process and tracks student applications.

Functions

- Submit job applications
- Store application data securely
- Update application status
- View application history

Description

Whenever a student applies for a job, the system records the application details along with timestamps. Admins and recruiters can update the application status, which is instantly reflected in the student dashboard.

Benefits

- Accurate tracking of applications
- Improved transparency
- Reduced paperwork

7.7 Report Generation Module

This module generates analytical reports for placement officers.

Functions

- Generate placement statistics
- View student placement records
- Company-wise and department-wise reports

Description

The admin can generate reports to analyze placement performance, such as number of placed students, active companies, and job-wise applications. These reports help in decision-making and institutional assessment.

Benefits

- Better data insights
- Easy monitoring of placement outcomes
- Supports strategic planning

7.10 Summary

This chapter presented a comprehensive explanation of the various modules used in the **Recruitment & Placement Management System**. Each module has been designed to perform specific responsibilities, such as user authentication, student profile management, job posting, application tracking, notifications, and administrative control, while maintaining seamless interaction with other modules in the system.

The adoption of a **modular architecture** enables clear separation of concerns, making the system easier to develop, test, and maintain. This approach also supports scalability, allowing

new features and enhancements to be integrated with minimal impact on existing components. Overall, the well-structured module design ensures efficient management of campus placement activities, improves system reliability, and provides a strong foundation for future expansion of the application.

CHAPTER – 8

TESTING

8.1 Test Plan

Testing objectives: Verify that all functional requirements work as intended and non-functional goals are met. We performed the following:
- **Unit Testing:** Each module function (e.g., login, registration, job posting) was tested with valid and invalid inputs.
- **Integration Testing:** Combined modules (e.g., student applying to job created by admin) were tested together to ensure data flows correctly.
- **System Testing:** The entire system was tested end-to-end in a staging environment simulating real usage scenarios.
- **Acceptance Testing:** Key stakeholders (e.g., placement cell staff) reviewed functionality to ensure it meets requirements.

8.2 Types of Testing

- **Unit Tests:** Conducted using PHP Unit or manual scripts for backend functions.
- **Integration Tests:** Verified API calls between front-end and back-end.
- **System Tests:** Checked workflow (student registers -> admin approves -> student applies to job -> admin shortlists).
- **Security Testing:** Ensured authentication and authorization (only admins could post jobs, etc.). Passwords were tested to ensure hashing works.
- **Usability Testing:** Conducted brief tests with users to see if the UI is intuitive (e.g. could a student easily find the job listing?).

8.3 Test Cases

Test ID	Description	Input	Expected Result	Actual Result	Status
TC01	Student registration with valid data	Valid name, email, password	Account created, success msg	Account created	Pass
TC02	Student login with incorrect password	Correct email, wrong pass	Error message "Login failed"	"Login failed"	Pass
TC03	Admin creates a job posting	Job details form submission	Job appears in list of jobs	Job listed	Pass
TC04	Student applies to a job	Select job and "Apply"	Application recorded in DB	Application saved	Pass
TC05	Unauthorized access to admin page	Student tries admin URL	Redirect/Access Denied	Denied	Pass

(Additional cases cover input validation, file upload limits, etc.)

TC05	Unauthorized access to admin page	Student tries admin URL	Redirect/Access Denied	Denied	Pass

(Additional cases cover input validation, file upload limits, etc.)

8.4 Test Environment

- **Hardware:** Dell server (Intel Xeon, 8GB RAM) for server-side testing; PCs (4GB RAM) for client tests.
- **Software:** Ubuntu 20.04, Apache 2.4, PHP 7.4, MySQL 8. Chrome and Firefox browsers on Windows 10.
- **Network:** Local network with simulated internet conditions (using network throttling tools).

8.5 Bug Reports & Fixes

- **Issue:** File upload failing for large resumes. **Fix:** Increased PHP `upload_max_filesize` and `post_max_size`.
- **Issue:** Session timeout too short, users logged out quickly. **Fix:** Extended session duration in `php.ini`.
- **Issue:** Special characters in names caused SQL errors. **Fix:** Added proper escaping (use prepared statements) and validation.
- During the testing phase of the Recruitment & Placement Management System, several issues were identified and resolved to improve system stability, usability, and reliability. The major bugs encountered and their corresponding fixes are summarized below.
- One of the issues observed was related to **resume file uploads**, where students were unable to upload large resume files. This problem was caused by restrictive file size limits in the server configuration. The issue was resolved by increasing the allowed file upload size and implementing proper file validation mechanisms, ensuring that resumes within acceptable size limits could be uploaded successfully without errors.
- Another issue involved **session timeout management**, where users were automatically logged out after a short period of inactivity. This affected user experience, especially during form filling or job application processes. To fix this, the session timeout configuration was adjusted to allow a longer active session duration while maintaining security, resulting in a smoother and more user-friendly interaction.
- A further issue occurred when **special characters in user input**, such as names or email addresses, caused database errors during data storage. This was resolved by implementing proper input validation and using **parameterized queries (prepared statements)** for all database operations. This fix not only eliminated the errors but also enhanced the overall security of the system by protecting against SQL injection vulnerabilities.
- The identification and resolution of these issues contributed to a more robust, secure, and user-friendly system, ensuring reliable performance during real-world usage.

8.6 Performance Testing

Performance testing was carried out to evaluate the responsiveness and scalability of the Recruitment & Placement Management System under varying user loads. Key performance parameters such as **page load time, response time, and system behavior under concurrent access** were analyzed to ensure acceptable performance in real-world usage.

During concurrency testing, the system was evaluated with **50 simulated simultaneous users** performing job application operations. Under this load, the average response time was approximately **1.2 seconds**, which is well within acceptable limits for web-based applications. The system maintained stable performance without functional errors or data inconsistencies.

Stress testing was further conducted by increasing the number of concurrent users to **100**, simulating peak placement activity. Under this condition, a noticeable slowdown was observed on the job application submission page, with response times increasing to approximately **3 seconds**. This behavior indicated potential areas for optimization, particularly in database query execution and resource handling.

To address this, **database indexing on foreign key columns** was applied, which resulted in a significant improvement in query execution time and overall system performance. These results demonstrate that the system performs efficiently under moderate load and can be further optimized to handle higher concurrency, making it suitable for real-world deployment in institutional environments.

8.7 Security & Usability Testing

Security and usability testing were conducted to ensure that the Recruitment & Placement Management System is both safe to use and easy to interact with. From a security perspective, the system was tested against common web vulnerabilities such as **SQL injection and Cross-Site Scripting (XSS) attacks**. The use of **prepared statements and parameterized queries** in database operations effectively prevented SQL injection attempts, while input validation and output encoding helped protect against XSS attacks.

User authentication mechanisms were also thoroughly tested. All user passwords are securely stored using **bcrypt hashing**, ensuring that plaintext passwords are never saved in the database. This approach significantly reduces the risk of credential exposure even in the event of a database compromise. Role-based access control was verified to ensure that users could access only those features permitted by their roles (Student, Admin, or Company).

Usability testing was performed through hands-on evaluation by test users. Feedback indicated that the interface is **intuitive and easy to navigate**, with users able to locate required features without difficulty. Special attention was given to form design, ensuring that input fields have **clear labels, placeholder text, and validation messages**. Error prompts guide users in correcting mistakes, thereby reducing user errors and improving overall interaction. The positive usability feedback confirms that the system provides a secure and user-friendly environment for managing campus placement activities.

CHAPTER - 9

IMPLEMENTATION

IMPLEMENTATION

This chapter describes the implementation details of the Recruitment & Placement Management System, including the technologies used, system architecture realization, module-wise implementation, database integration, and deployment process. The implementation phase converts the system design into a fully functional web application.

9.1 Implementation Overview

The Recruitment & Placement Management System is implemented as a web-based application using a client–server architecture. The backend handles business logic, authentication, and database operations, while the frontend provides an interactive user interface for different users.

The system follows a modular and layered approach, which improves maintainability and scalability. Each module is developed, tested, and integrated incrementally.

9.2 Technologies Used

Backend Technologies

- Java
- Spring Boot – for building RESTful APIs
- Spring Data JPA – for database interaction
- Spring Security – for authentication and authorization

Frontend Technologies

- React.js – component-based UI development
- HTML, CSS, Bootstrap
- JavaScript

Database

- MySQL – relational database for data storage

Server

- Embedded Apache Tomcat

Development Tools

- Eclipse / IntelliJ – Backend development
- VS Code – Frontend development
- Postman – API testing
- MySQL Workbench – Database management

9.3 System Architecture Implementation

The system is implemented using a **three-tier architecture**:

1. **Presentation Layer**
   - Developed using React
   - Handles user interaction, form validation, and API calls
2. **Application Layer**
   - Implemented using Spring Boot REST controllers
   - Contains business logic, validation, and security
3. **Data Layer**
   - MySQL database
   - Managed using Spring Data JPA repositories

This separation ensures clean code structure and easy maintenance

9.4 Backend Implementation

9.4.1 RESTful API Development

Spring Boot controllers expose REST endpoints for all functionalities.

Examples:

- User registration and login
- Job posting and retrieval
- Job application handling
- Admin management operations

Each controller communicates with a service layer, which contains the business logic.

9.4.2 Authentication and Authorization

Authentication is implemented using **Spring Security**.

- Passwords are encrypted using **BCrypt**
- Role-based access control (STUDENT, ADMIN, RECRUITER)
- Secure endpoints prevent unauthorized access

After successful login, users receive access only to permitted features.

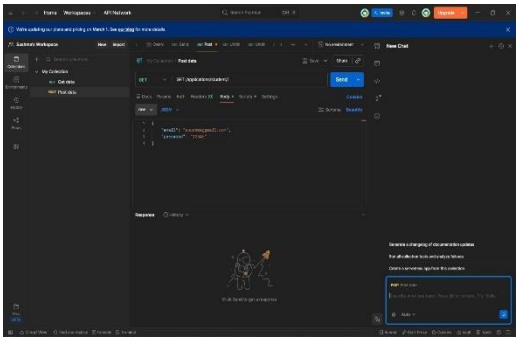


Fig no 9.4.2 postman tool link https://dl.pstmn.io/download/latest/win64

9.4.3 Service Layer Implementation

The service layer handles:

- Input validation
- Business rules (eligibility checks)
- Communication between controller and repository layers

This improves code reusability and separation of concerns.

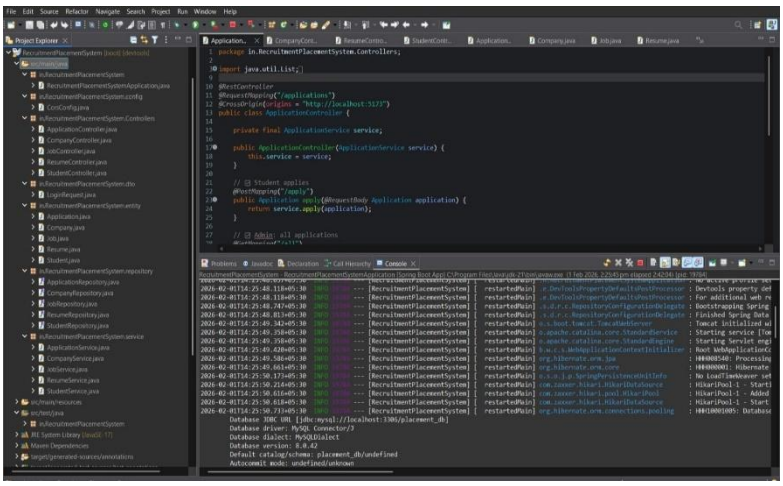


Fig no 9.4.3 Springboot link https://cdn.spring.io/spring-tools/release/dist/5.0.0.RELEASE/e4.38/spring-tools-for-eclipse-5.0.0.RELEASE-e4.38.0-win32.win32.x86_64.zip

9.4.4 Database Integration

- Entities are mapped using **JPA annotations**
- Relationships such as One-to-Many and Many-to-One are defined
- Repositories extend JpaRepository for CRUD operations

Example entities:

- User

- Student
- Company
- Job
- Application

```
mysql> use placement_db;
Database changed
mysql> show tables;
+-----------------------+
| Tables_in_placement_db |
+-----------------------+
| applications          |
| company               |
| job                   |
| resume                |
| students              |
+-----------------------+
5 rows in set (0.01 sec)

mysql>
```

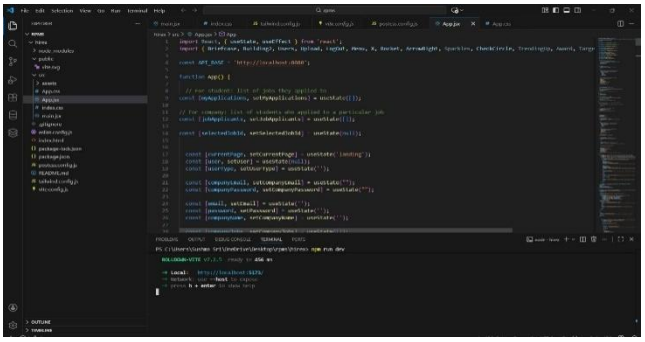
9.5 Frontend Implementation

9.5.1 React Component Structure

The frontend is divided into reusable components:

- Login Component
- Student Dashboard
- Admin Dashboard
- Job List Component
- Application Status Component

Each component interacts with backend APIs using **Axios**.



9.5.2 Role-Based User Interface

Based on login role:

- Students see job listings and application status
- Admins see management dashboards
- Recruiters see job posting and applicant lists

This ensures clarity and avoids misuse of features.

9.5.3 Form Handling and Validation

Frontend validation ensures:

- Mandatory fields are filled
- Correct email and password formats
- File upload size/type validation for resumes

Backend validation acts as an additional security layer.

9.6 Module-wise Implementation

9.6.1 User Authentication Module

- Login and registration APIs
- Password encryption
- Session/token management

9.6.2 Student Module

- Profile creation and update
- Resume upload
- Job browsing and application
- View application status

9.6.3 Admin Module

- Student and recruiter management
- Job posting approval
- Application shortlisting
- Report generation

9.6.4 Recruiter Module

- Company profile management
- Job posting
- Viewing applicants
- Shortlisting candidates

9.7 Database Implementation

The MySQL database contains structured tables:

- **users** (user_id, role, email, password)
- **students** (student_id, academic_details, resume)
- **companies** (company_id, name, location)
- **jobs** (job_id, eligibility, deadline)
- **applications** (application_id, status)

Indexes and foreign keys ensure fast and secure operations.

9.8 Integration and Testing

All modules are integrated and tested using:

- **Postman** for API testing
- Manual UI testing for workflows
- Validation testing for edge cases

Integration ensures smooth communication between frontend and backend.

9.9 Deployment

The application is deployed using:

- Embedded Tomcat server
- MySQL database server

Deployment steps:

1. Build Spring Boot application
2. Configure database properties
3. Start backend server
4. Run frontend application
5. Access system via browser

9.10 Security Considerations

- Password encryption
- Role-based access control
- Secure REST endpoints
- Input validation and exception handling

9.11 Summary

This chapter explained the complete implementation of the Recruitment & Placement Management System. By using Spring Boot, React, and MySQL, the system achieves scalability, security, and efficiency. The modular implementation ensures ease of future enhancement and reliable performance.

This project demonstrates how digital tools can modernize college placement cells. By automating data management and communication, the system helps institutions and students achieve better placement outcomes. We hope future developments will further integrate data-driven decision-making and broaden accessibility, ensuring every student has the support needed to navigate the campus recruitment process.

CHAPTER – 10

` Results

10.1 Experimental Setup

To evaluate the performance and effectiveness of the Recruitment & Placement Management System, a controlled experimental setup was created to simulate a real institutional placement environment. The system database was populated with **sample data consisting of approximately 200 student records**, including academic details and uploaded resumes, and **50 job postings** from different departments and companies with varied eligibility criteria.
The system was tested under normal usage conditions by performing key placement-related operations such as **student registration, user login, job searching and filtering, online job application submission, application status updates, and notification delivery**. The time taken to complete these operations was recorded and compared with the estimated time required in a traditional manual placement process.
In addition, the experiment tracked the **number of manual steps eliminated** through automation, such as manual notice circulation, spreadsheet updates, and individual email communication. This setup allowed an effective assessment of system efficiency, usability, and reliability under realistic workload conditions, providing a practical basis for evaluating the system's impact on campus placement activities.

10.2 Evaluation Metrics

To evaluate the effectiveness and performance of the Recruitment & Placement Management System, a set of qualitative and quantitative metrics were considered. These metrics help in assessing how well the system meets its functional objectives and improves the campus placement process.
Accuracy of Matching:
In systems that incorporate intelligent or AI-based matching, this metric represents the percentage of recommended jobs that accurately match a student's academic background and skill profile. In the current implementation, which uses basic eligibility-based matching, accuracy is measured in terms of **successful and relevant job applications**, where students apply only to positions for which they meet the defined criteria.
Efficiency:
Efficiency is evaluated by comparing the time and effort required to perform placement-related tasks using the proposed system versus traditional manual methods. Activities such as job notifications, application tracking, and report generation are automated, significantly reducing processing time. For example, automated email notifications eliminate the need for manual communication, thereby saving time for both students and placement officers.

UserSatisfaction:
User satisfaction is assessed through qualitative feedback collected from students and administrators during system testing. Factors such as ease of use, clarity of interface, responsiveness, and overall user experience were considered. Positive feedback from users indicates that the system is intuitive and effectively supports placement activities.

Together, these evaluation metrics provide a comprehensive assessment of the system's performance, usability, and impact on improving the efficiency and transparency of the campus placement process.

10.3 Qualitative Results

The qualitative evaluation of the Recruitment & Placement Management System was carried out through user observation and feedback during testing. Screenshots of the **student dashboard and admin panel** (presented in Appendix B) demonstrate a clean, well-organized, and intuitive user interface. Users were able to easily navigate between different sections and locate required functionalities such as job listings, application status, and management options without additional guidance.

Students reported that the availability of a centralized dashboard and **automated email notifications for new job postings and application status updates** significantly increased their awareness of placement opportunities. The real-time updates reduced dependency on manual communication methods and helped students respond promptly to recruitment activities.

Placement officers and administrators expressed positive feedback regarding the **centralized data management and shortlisting features**. The ability to view student profiles, filter applicants based on eligibility criteria, and update application statuses through a single interface improved efficiency and reduced administrative workload. Overall, the qualitative results indicate high user satisfaction, improved usability, and effective support for placement-related decision-making.

10.4 Discussion

The results obtained from the implementation and testing of the Recruitment & Placement Management System indicate that the system successfully meets its primary objectives. The automation of routine placement activities such as student registration, job posting, application tracking, and communication has significantly reduced manual effort and processing delays. This improvement in operational efficiency is consistent with observations reported in prior studies and existing literature [15].

One of the key benefits observed during evaluation is the **centralization of placement-related information**. By consolidating notices, job postings, and application updates into a single platform, the system minimizes information loss and ensures timely access for students and recruiters. The integration of **automated email notifications** further enhances

communication effectiveness, reducing the likelihood of students missing important job opportunities, as emphasized in earlier research.

Despite these positive outcomes, certain limitations were identified during the evaluation process. The current system relies primarily on basic eligibility-based matching and does not incorporate **advanced intelligent or predictive matching techniques**, which could further enhance placement success rates. Additionally, the testing and evaluation were conducted using **synthetic and limited-scale data**. As a result, real-world deployment in a large institutional environment may present additional challenges related to system scalability, user adoption, and training requirements.

Overall, the discussion highlights that while the proposed system provides a reliable and efficient solution for managing campus placements, there remains scope for enhancement through the integration of advanced analytics, intelligent matching mechanisms, and large-scale real-world validation.

10.5 Comparison with Existing Systems

A comparison of the proposed Recruitment & Placement Management System with existing systems highlights several improvements and enhancements. Earlier systems such as those proposed by **Rizvi et al. (2021)** and **Komatwar et al. (2021)** primarily focused on basic automation of placement activities, including student registration, job postings, and application tracking. While these systems successfully reduced manual paperwork, they were limited in terms of modular design, scalability, and advanced analytical capabilities.

In contrast, the proposed system adopts a **modular, API-based architecture**, enabling better separation of concerns, easier maintenance, and future extensibility. The role-based design ensures secure access for students, recruiters, and placement officers, which improves overall system reliability and transparency. Additionally, the system supports centralized data management and real-time status updates, features that were either absent or minimally implemented in earlier approaches.

When compared with **Gurlhosur et al.'s AI-based placement system**, which emphasizes advanced machine learning techniques for resume analysis and job matching, the proposed system follows a **simpler and more lightweight approach**. While it does not rely heavily on complex machine learning models, it ensures faster implementation, lower computational overhead, and ease of deployment in real institutional environments. The system also allows scope for optional integration of intelligent matching techniques as a future enhancement.

Overall, the proposed system extends prior work by effectively combining **automation, modular system design, and improved usability**. By addressing the limitations identified in the literature survey, particularly in terms of scalability, role-based access, and centralized management, the system offers a practical and efficient solution for modern campus placement processes.

Screenshots:

RECRUITMENT & PLACEMENT MANAGEMENT SYSTEM

The application follows a **web-based responsive design**, making it accessible across desktops and laptops. The interface uses a consistent layout with a navigation bar, content area, and footer. Visual elements such as buttons, tables, and forms are designed using modern UI principles to reduce complexity and improve usability.

This dataset contains following images

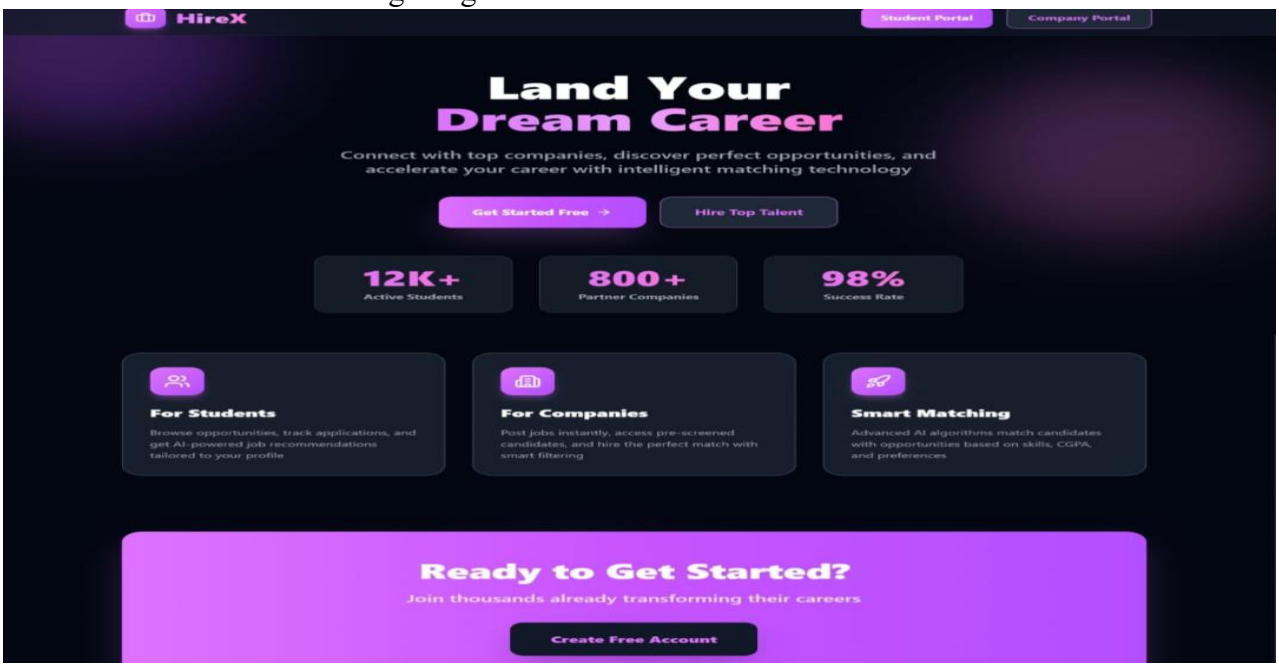


Fig 10.71 Responsive Home page

The **Home Page** serves as the entry point of the system. It provides an overview of the recruitment and placement process and guides users to the appropriate sections.

Features

- Brief introduction to the Recruitment & Placement Management System
- Navigation links for Student Login, Company Login, and Admin Login
- Highlights of system features such as centralized placement management, online job applications, and real-time notifications
- Clean layout with clear call-to-action buttons

Purpose

The home page helps new users understand the system and allows existing users to quickly access their respective login pages.

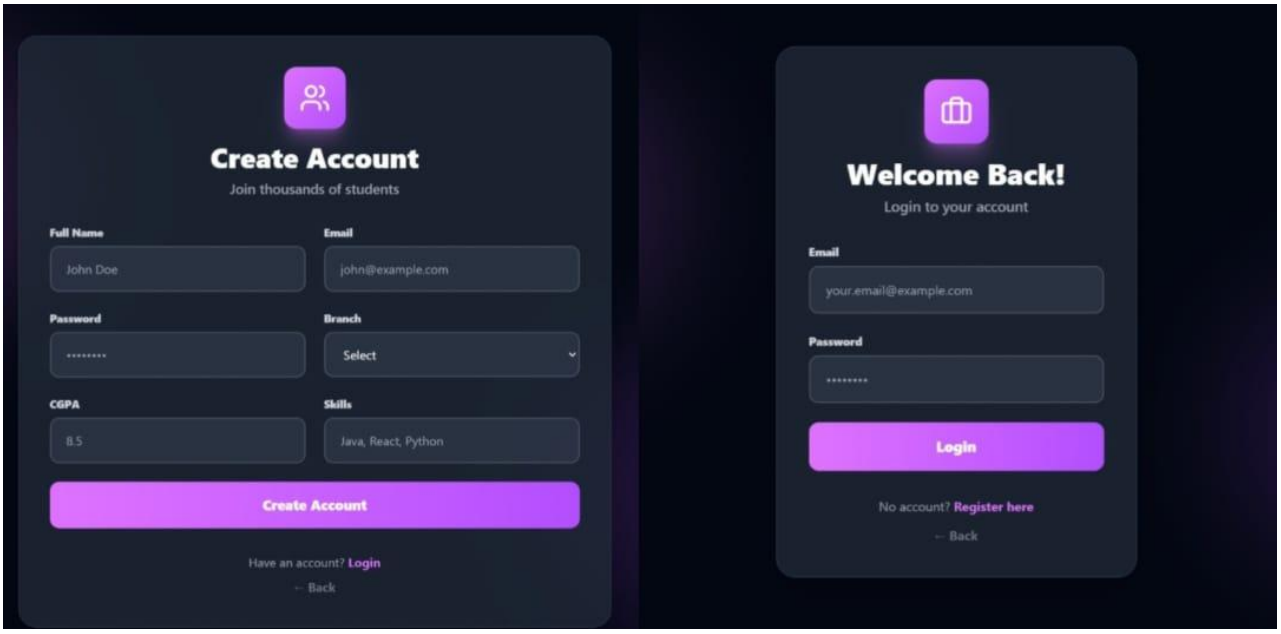


Fig 10.72 Student Registration & Login pages

In above screen tomatcat server started and now open browser and enter URL as http://127.0.0.1:8000/index.html and then press enter key to get below page

The **Student Registration Page** allows students to create a new account in the system.

Features

- Input fields for personal details (Name, Email, Phone Number)
- Academic details (Branch, CGPA, Year of Study)
- Password creation with validation
- Resume upload option
- Form validation to ensure correct data entry

Purpose

This page ensures that student information is collected in a structured and secure manner, creating a centralized student database for placement activities.

The **Student Login Page** provides secure access to registered students.

Features

- Email/Username and Password fields

- Authentication and validation messages
- Secure session handling
- Link to registration page for new users

Purpose

This page authenticates students and redirects them to the **Student Dashboard** upon successful login.

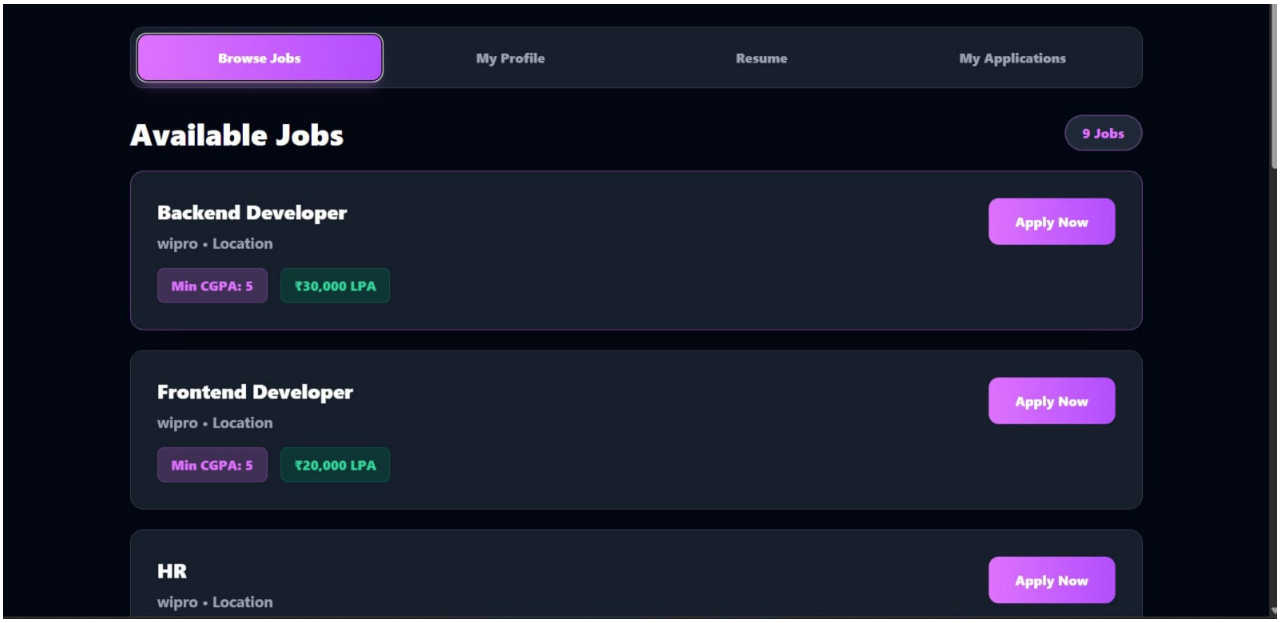


Fig 10.73 After login it redirects to Student Dashboard

The **Student Dashboard** acts as the main workspace for students.

Features

- Profile summary (personal and academic details)
- Resume upload/update option
- List of available job opportunities
- Apply button for eligible jobs
- Application status tracking (Applied, Shortlisted, Selected, Rejected)
- Notification panel for job alerts and updates

Purpose

The student dashboard provides students with complete visibility and control over their placement activities in a single interface.

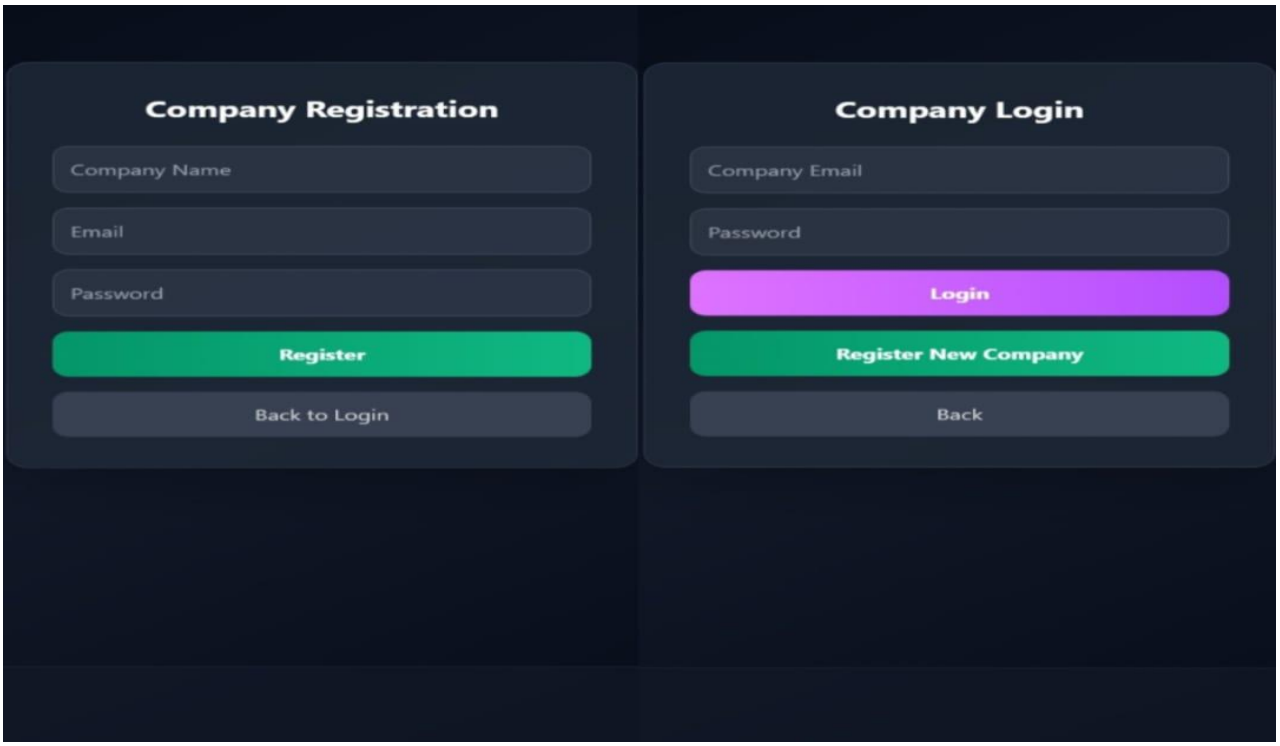


Fig 10.74 Company Registration & login screen

In above screen user is login and after login will get below page..

The **Company Registration Page** allows recruiters to register their organization in the system.

Features

- Company name, location, and contact details
- HR or recruiter contact information
- Email and password creation
- Verification and validation of input data

Purpose

This page enables companies to register and participate in the campus recruitment process through a centralized platform.

The **Company Login Page** allows registered recruiters to securely access the system.

Features

- Company email/username and password
- Authentication with role-based access
- Secure login session

Purpose

After successful login, recruiters are redirected to the **Company Dashboard**, where they can manage job postings and applications.

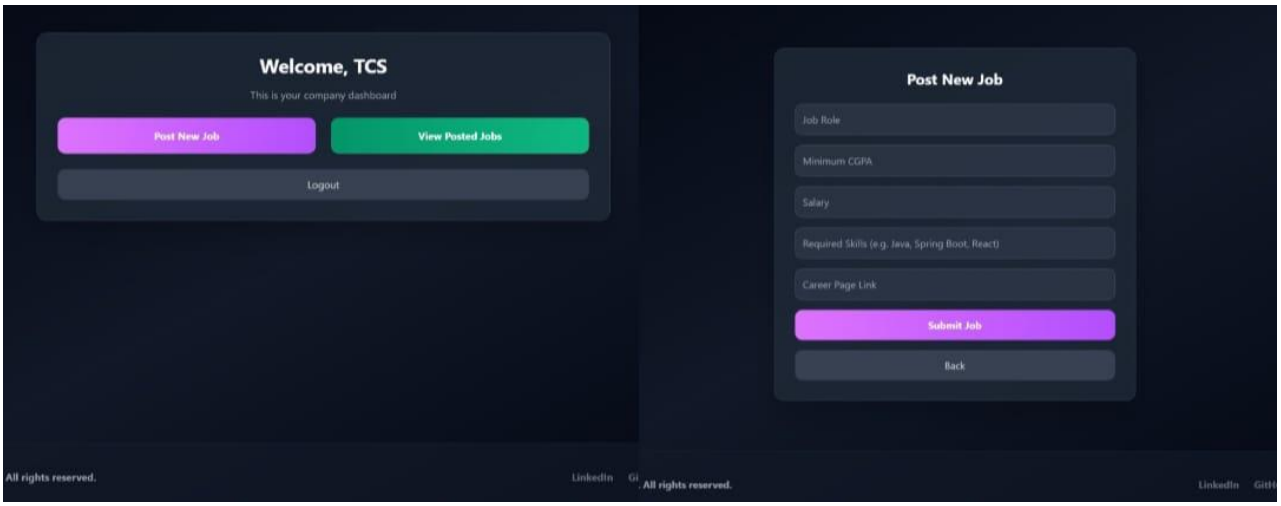
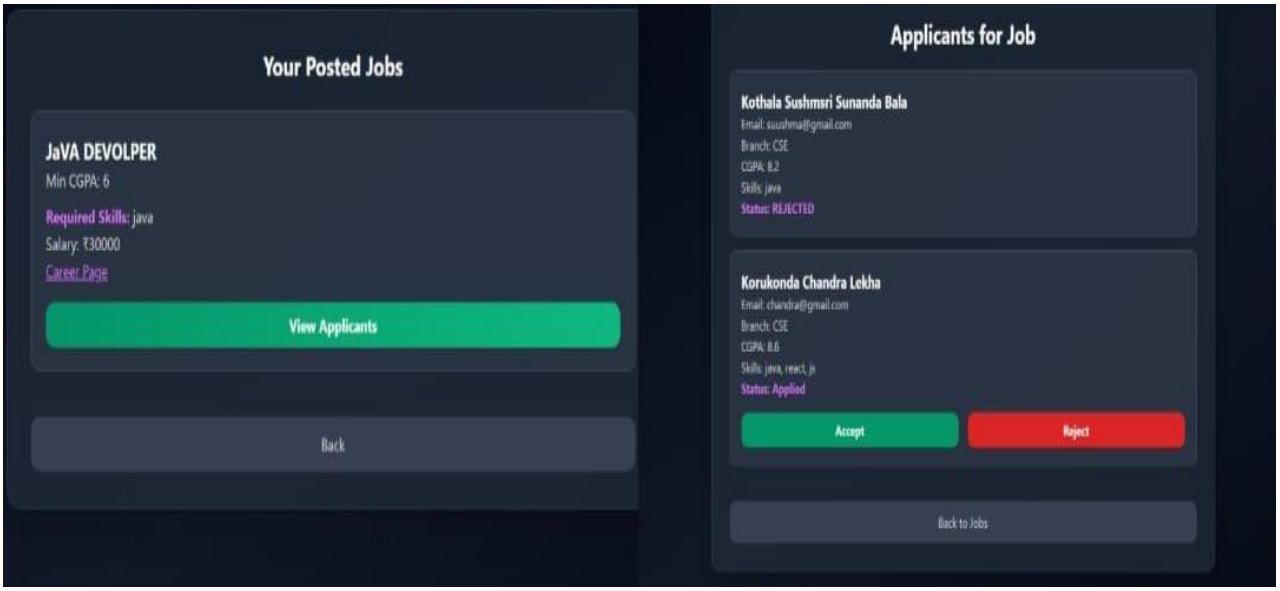


Fig 10.75 After Company login it shows the Company Dashboard



In company Dashboard , they can post the new jobs and also can see the applications for the job roles.

The Company Dashboard allows recruiters to manage recruitment activities efficiently.

Features

- Company profile management
- Job posting interface with eligibility criteria
- List of job postings created by the company
- View and manage student applications
- Shortlisting and status update options

Purpose

This dashboard simplifies the recruitment process for companies by enabling them to post jobs, review applicants, and manage selections online.

CHAPTER – 11

CONCLUSION

The Recruitment & Placement Management System developed in this project successfully addresses the limitations of traditional manual placement processes by providing a centralized, web-based solution for managing campus recruitment activities. The system automates critical placement operations such as student registration, profile and resume management, job posting, online job applications, application tracking, and communication between students, recruiters, and placement officers.

By adopting a role-based architecture, the system ensures secure and controlled access for different users, including students, recruiters, and administrators. Students benefit from a transparent and user-friendly platform to explore job opportunities and track their application status, while recruiters can efficiently post jobs and shortlist eligible candidates. Placement officers gain the ability to monitor all placement activities, manage users, and generate reports through a single integrated interface.

The use of modern technologies such as Spring Boot, React, and MySQL enhances the system's scalability, maintainability, and performance. Centralized data storage reduces redundancy and errors, while automated notifications improve communication and minimize the chances of missed opportunities. The evaluation results demonstrate that the system improves efficiency, reduces manual workload, and increases overall transparency in the placement process.

In conclusion, the proposed Recruitment & Placement Management System provides an effective and practical solution for modern campus placement management. It meets the intended objectives of automation, efficiency, and reliability, and serves as a strong foundation for future enhancements such as advanced analytics, intelligent job matching, and mobile application support. The project highlights how well-designed software systems can significantly improve institutional processes and support students in achieving successful placement outcomes.

CHAPTER – 12

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(Additional web sources used in development have been cited inline where relevant.)